
Mateus Bonette

Software & Applied AI Engineer

Python, FastAPI, RAG, LLMs, AI Agents

Itajubá, MG, Brazil

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PROFESSIONAL SUMMARY

Software & Applied AI Engineer with 4+ years of cumulative experience building web applications, backend systems, and production AI solutions. Hands-on experience with Python, FastAPI, RAG, LLMs, and AI agents, including document-processing pipelines, Human-in-the-Loop workflows, and observability. Co-founder of a financial SaaS product with 100+ users and experience with cloud architectures on Google Cloud and AWS.

PROFESSIONAL EXPERIENCE

Co-founder - Software & AI Engineer

Junta Já - Remote - juntaja.com.br - Jun 2026 - Present

- Lead engineering for a production financial SaaS used by 100+ users, building backend services and AI pipelines for invoice and bank-statement processing.
- Automated document classification, structured extraction, normalization, and categorization with Python, FastAPI, Gemini, and Vertex AI, reducing manual steps before human review.
- Designed Human-in-the-Loop validation, deterministic rules, Tool Calling, and MCP workflows before PostgreSQL persistence, improving traceability and processing safety.
- Integrated WhatsApp and Mercado Pago and deployed services on Google Cloud with Cloud Run, Cloud SQL, Cloud Storage, Secret Manager, and CI/CD, connecting document ingestion, billing, and product operations.

Full Stack Developer - Applied AI, RAG & LLMs

Moontech - Remote - May 2024 - Jun 2026

- Automated three operational workflows with LLMs integrated into HubSpot - lead qualification, support, and post-sales - reducing manual work and saving hours each week.
- Designed RAG pipelines with LangChain, LangGraph, embeddings, chunking, and PostgreSQL/pgvector for contextual retrieval in AI applications.
- Instrumented observability with Langfuse to monitor prompts, tokens, latency, and costs, supporting diagnosis and continuous optimization of LLM workflows.
- Delivered internal and web applications with Python, FastAPI, Angular, Next.js, GraphQL, PostgreSQL, Streamlit, Docker, and AWS in production environments.

Full Stack Developer

byron.solutions - UNIFEI Junior Enterprise - Itajubá, MG / Remote - Apr 2020 - Nov 2021

- Delivered websites, landing pages, and web systems for clients, including backend services, responsive interfaces, CRUD functionality, REST APIs, and automations.
- Led requirements gathering, scope definition, prototyping, and technical deliveries in direct collaboration with clients and stakeholders.

TECHNICAL SKILLS

AI & LLM Engineering: LLMs, RAG, AI Agents, LangChain, LangGraph, Prompt Engineering, Tool Calling, MCP, Embeddings, Retrieval, Human-in-the-Loop, Langfuse, Evals, Guardrails

Backend & Data: Python, FastAPI, REST APIs, PostgreSQL, pgvector, SQLAlchemy, Node.js, GraphQL, Data Pipelines

Cloud & DevOps: Google Cloud, AWS, Docker, Git, GitHub, GitHub Actions, CI/CD, Linux

Frontend & Data: TypeScript, JavaScript, React, Angular, Next.js, Pandas, NumPy, Scikit-learn, Streamlit

LANGUAGES

Portuguese: Native. **English:** Advanced technical reading and writing; intermediate-to-advanced speaking.

SELECTED PROJECTS

Ponte Viva - Applied AI social platform. Connects organizations, volunteers, and local needs through contextual search and recommendations powered by RAG and AI agents. Architecture with FastAPI, PostgreSQL/pgvector, LangChain, LangGraph, embeddings, BM25, hybrid retrieval, reranking, evals, Langfuse, and guardrails.

CoinSight - Cryptocurrency market prediction and analysis. Undergraduate thesis focused on analysis and prediction using market data, news, and external factors with Python, Machine Learning, Streamlit, and PostgreSQL. Repository: [coinsight_tcc](https://github.com/mateus-bonette00/coinsight_tcc)

FBA Automation - Full Stack automation for e-commerce. System for product research and supplier analysis, reducing manual tasks with React, FastAPI, Python, Playwright, PostgreSQL, and Docker. Repository: [fba-automation](https://github.com/mateus-bonette00/fba-automation)

EDUCATION

Federal University of Itajubá (UNIFEI) - B.Sc. in Computer Science - 2020 - 2025